

# HERMES: Four-Layer Joint Optimization Architecture

*colibri Engine (glm.c) — #ifdef HERMES hooks*

## L1: Scheduler

Low-rank Predictor → PoE Fusion → Online Learning

Hook 1

## L2: Compression

Per-domain K-SVD Dictionary + LZ4

Hook 2

## L3: Precision

Sensitivity Profiling → Dynamic int8/int4/int2

Hook 2

## L4: Vulkan Compute

SPIR-V int4/int8 GEMM → Timeline Semaphore Pipeline

Hook 3

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Multiplicative  
Coupling

SSD/NVMe

→ RAM (Hot/Warm Cache)

→

VRAM

$$S_{\max} = \min((1 + 1/\alpha) c_1 c_2, (\alpha + 1) p g)$$